Feature List with Ruthless Prioritization

MoSCoW method: every Must is shown live in the demo; nothing else is started until all Musts work end-to-end

Team Roles (6 members)

Code	Role	Primary responsibility
ML	ML / Downscaling Lead	Temperature + rainfall models, validation metrics
DE	Data Engineer	Data download, regridding, feature layers, panchayat zonal stats
BE	Backend Developer	FastAPI, PostGIS schema, advisory rule engine, scheduler
FE1	Frontend Developer (Maps)	Leaflet split map, panchayat detail, validation charts
FE2	Frontend Developer (UX)	Advisory, share & farmer mobile screens, responsiveness, i18n
PM	Product & Pitch Lead	Advisory rules content, translations, demo script, slides, testing

MoSCoW Feature Table

Priority	Feature	Owner	Status	Notes
MUST	**M1 Panchayat Downscaling Engine**	ML, DE	Planned (H0-H16)	Hybrid: lapse-rate + XGBoost residual for Tmax/Tmin; occurrence classifier + amount regressor + quantile mapping for rain; RH/wind derived. Mass-consistent with block forecast. **Demo step 1-3.**
MUST	**M2 Block vs Panchayat Split Map**	FE1, BE	Planned (H8-H22)	Side-by-side Leaflet map, variable + day toggles, colour legend. The single most important visual. **Demo step 2.**
MUST	**M3 Crop Advisory Rule Engine**	BE, PM	Planned (H10-H24)	YAML rules for 3 crops (cotton, paddy, maize) x stages: spray, irrigate, sow, harvest, fertilizer. Ranked by urgency. **Demo step 4.**
MUST	**M4 Confidence Badge + Validation View**	ML, FE1	Planned (H14-H28)	High/Med/Low from model spread; RMSE, bias, POD/FAR/CSI vs block baseline at held-out AWS; one backtest event day. **Demo steps 3 & 6.**
MUST	**M5 Local-Language Advisory & Share**	FE2, PM	Planned (H18-H30)	Telugu / Hindi / English templates; SMS + WhatsApp preview on phone mockup; simulated send log. **Demo step 5.**
SHOULD	**S1 Monsoon Onset & False-Onset Alert**	ML, BE	Backlog	Agronomic onset per panchayat from downscaled rain + historical climatology; "don't sow yet" warning. Synergy with SIH 26086.
SHOULD	**S2 Extreme-Weather Alerts**	BE	Backlog	Heatwave, heavy rain, strong wind thresholds per panchayat; red banner on map.
SHOULD	**S3 Officer Bulk Dispatch Dashboard**	FE2, BE	Backlog	Select multiple GPs, approve advisories, see delivery log.
SHOULD	**S4 Downloadable Advisory Bulletin (PDF)**	BE	Backlog	GKMS-style bulletin per block with panchayat table.
SHOULD	**S5 Farmer Feedback ("Did it rain today?")**	FE2	Backlog	One-tap feedback stored for future validation; crowdsourced ground truth.
COULD	**C1 IVR / voice advisory**	PM	Idea	Text-to-speech Telugu audio for low-literacy users.
COULD	**C2 U-Net super-resolution comparison**	ML	Idea	Deep-learning benchmark vs hybrid model; stretch only.
COULD	**C3 Soil-moisture water balance**	ML	Idea	Improves irrigation advice using ERA5-Land / SMAP.
COULD	**C4 Offline PWA for farmers**	FE2	Idea	Cache last forecast on the phone.
COULD	**C5 Farmer crop profile personalisation**	FE2, BE	Idea	Advisory tailored to each farmer's plot and sowing date.

Priority	Feature	Owner	Status	Notes
WON'T	**W1 Own NWP / WRF model runs**	-	Out	Too compute-heavy for 36 h; we post-process IMD output instead.
WON'T	**W2 Nationwide coverage**	-	Out	One pilot district; architecture is designed to scale.
WON'T	**W3 Real bulk SMS gateway**	-	Out	Simulated / sandbox send only.
WON'T	**W4 Crop yield prediction**	-	Out	Different problem; not needed for advisories.

Prioritization Rules We Follow

- **Maximum 5 Musts** - each one maps to a numbered step in the live demo (see Doc 02 and Doc 06).
- **Vertical slice first:** by hour 20, one panchayat must work end-to-end (data -> model -> API -> map -> advisory) even with rough UI. Polish comes after.
- **Shoulds start only when all Musts pass the demo dry-run.** S1 (monsoon onset) is the first Should to pick up because it reuses the rainfall output.
- **Could / Won't** items go on the "Future Scope" slide, not into the code.
- Any feature that cannot be shown working in under 15 seconds is cut from the demo.

36-Hour Build Timeline

Hours	Milestone	Exit criteria
H0-H6	Data ready	DEM, LULC, climatology, boundaries, sample forecast on a common 1 km grid
H6-H16	Models trained	Tmax/Tmin + rain models trained; metrics computed on held-out AWS
H16-H22	Vertical slice	API returns panchayat forecast; map renders; 1 advisory works end-to-end
H22-H30	All Musts complete	M1-M5 working together on the pilot district
H30-H33	Polish + Shoulds	UI polish, Demo Mode cache, S1 if time allows
H33-H36	Rehearse	3 full demo run-throughs; judge Q&A practice; backup video recorded